

Job Recruitment Portal — Software Design Document (SDD)

Version: 1.0

1. Architecture

JRP uses a three-tier web architecture: responsive browser UI, REST-style application/API layer, and relational database. Authentication and authorization are enforced in the application layer. An email adapter is isolated behind a notification service.

2. Major Components

Component	Responsibility	Requirements
Authentication & RBAC	Registration, login, role authorization	FR-CAN-001/002, FR-ADM-002, NFR-001
Candidate Profile	Candidate profile maintenance	FR-CAN-003
Job Service	Create/manage/search/view jobs	FR-CAN-004/005, FR-REC-001/002
Application Service	Submit/list applications and enforce business rules	FR-CAN-006/007, BR-002/003
Recruiter Review	Applicant list/details/status workflow	FR-REC-003/004/005
Administration	Account status, roles, audit viewer	FR-ADM-001/002/003
Audit Service	Immutable business/security event records	NFR-005
Notification Adapter	Sends transactional emails where configured	Supporting

3. Data Design

Entity	Key Fields	Relationships
User	user_id, email, password_hash, role, active	1:1 CandidateProfile where applicable
CandidateProfile	candidate_id, name, phone, skills, resume_ref	belongs to User
Job	job_id, recruiter_id, title, description, location, status, closing_date	belongs to recruiter; has Applications
Application	application_id, job_id, candidate_id, status, submitted_at	belongs to Job and Candidate
AuditEvent	event_id, actor_id, action, entity_type, entity_id, timestamp	references actor/entity

Unique constraint: `(candidate\_id, job\_id)` on Application enforces BR-002.

4. Interface Design

Endpoint	Method	Purpose
/api/auth/register	POST	Candidate registration
/api/auth/login	POST	Authenticate user

/api/profile	GET/PUT	Read/update candidate profile
/api/jobs	GET/POST	Search jobs / recruiter creates job
/api/jobs/{id}	GET/PUT	View/manage job
/api/jobs/{id}/applications	POST/GET	Candidate applies / recruiter lists applicants
/api/applications	GET	Candidate lists own applications
/api/applications/{id}/status	PATCH	Recruiter changes status

5. Security Design

- Server-side RBAC checks on every protected endpoint.
- Password hashing, secure session/token handling, input validation, parameterized database access.
- Candidate PII returned only to authorized recruiter/admin contexts.
- Audit events for administrative changes and application status transitions.

6. Error Handling

API errors return a stable code and user-safe message. Validation failures return field-level guidance. Internal details are logged server-side and not exposed to end users.

7. CR-001 Design Impact (Proposed, not baseline)

If approved, Application Service gains a retract operation, Application status gains `Retracted`, an audit event is recorded, candidate UI exposes Retract only for eligible statuses, and recruiter views display the retracted state. Proposed endpoint: `POST /api/applications/{id}/retract`.